

Vaibhav Sethi

Data Analyst | Relocating to Toronto, ON | Available full time January 2027
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EXPERIENCE

Junior Data Analyst, BrookSphere Research

May 2024 - Sep 2024, May 2025 - Sep 2025

Small market research firm running custom B2B and B2C surveys

- Cleaned, coded and analyzed client survey data in SPSS across concurrent B2B and B2C studies.
- Automated repeat cleaning and tabulation steps in R and Python, cutting turnaround on recurring studies.
- Built the tables and charts that broke survey results down by customer group for client reports.
- Handled client data requests and revisions against tight deadlines; asked back for a second summer.

Printing Services Representative, Staples Canada

Dec 2022 - Jun 2023

Office supply and print retailer

Vancouver, BC

- Completed 50 to 70 print orders per shift against fixed deadlines and fixed equipment problems on the spot.

PROJECTS

Mortgage Portfolio Stress-Testing Engine

[Live dashboard](#) | [Case study](#)

Python, SQL, DuckDB, R, SAS, XGBoost, Power BI

- Projected 13-quarter losses on a \$35.4B book of 147,609 US mortgages under Federal Reserve 2026 stress scenarios: \$65M in the baseline case against \$352M in the severe case.
- Built the SQL and DuckDB pipeline over 19 million monthly records for 350,000 real Freddie Mac loans.
- Trained a default model and tested it on loan years it had never seen (Gini 0.66, KS 0.50); calibrated its scores into real probabilities, cutting probability error 63% (Brier 0.175 to 0.064).
- Shipped a live dashboard with a plain English write-up, plus SAS and R versions and a Power BI export.

Credit Scoring and Expected Loss App

[Live scorecard](#)

R, Shiny, XGBoost, SHAP

- Built a web app that predicts which credit card clients will default and gives plain reasons for each score.
- Scored 6,001 held-out clients from public UCI data (AUC 0.77); added bulk CSV scoring for non-technical users.

LIGO Gravitational-Wave Glitch Classifier

Python, scikit-learn, Random Forest

- Sorted 7,966 detector noise samples into 22 types (largest outnumbered smallest 67 to 1), reaching 88% accuracy and 0.81 macro F1 (which weights rare types equally) on unseen data.

SKILLS

Programming and Databases: SQL (joins, CTEs, window functions), Python (pandas, NumPy, scikit-learn, XGBoost), R (tidyverse, ggplot2, Shiny), SAS, SPSS, DuckDB, Excel

Statistics and Modeling: Statistical analysis, regression, logistic regression, generalized linear models, time series, clustering, PCA, survey analysis, cross-tabulation, machine learning, model validation (AUC, Gini, KS), probability calibration, credit risk (PD, expected loss), stress testing

Reporting and Dashboards: Power BI, Tableau, Shiny, data visualization, Excel reporting, written reports for non-technical stakeholders

Data and Tools: ETL and data pipelines, data cleaning, feature engineering, SHAP, Git, Jupyter, RStudio, Parquet

EDUCATION

University of British Columbia, Vancouver, BC

Coursework complete 2026, degree 2027

B.Sc., Combined Major in Science: Statistics, Physics, Earth and Environmental Sciences

Certification: Bloomberg Market Concepts (BMC)